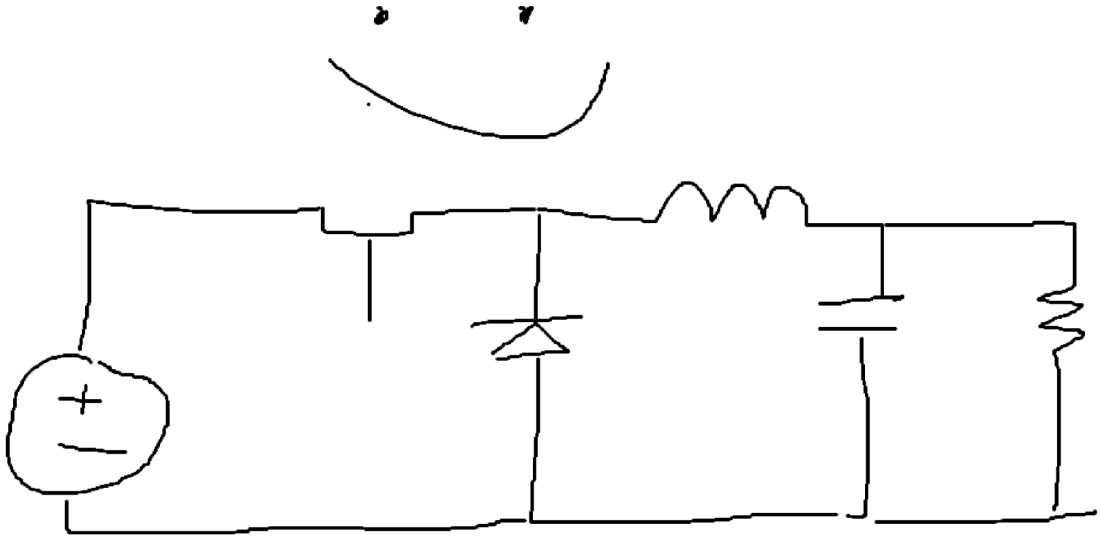
Formula Slug – Buck Converter Schematic Block
Power Electronics Onboarding Project 2026-2027

Overview

This project is centered around designing a buck converter to help power different boards on our LV system. This project requires you to design your own buck converter, sourcing components and creating the schematic and layout. You will also push it onto FS's github.



Our Low Voltage system consists of all our important logic and sensing boards, such as our Vehicle Control Unit, Telemetry boards, and Tractive Battery board. On all these boards, there are a variety of different components, who all may need to be powered by a different voltage level. For example, relays can operate at 12V, MCUs can operate at 5V, while some smaller chips run on as small as 3.3V. This is a big issue, because our low voltage bus runs on 12V!? If we overvolted those components, they would immediately stop working.

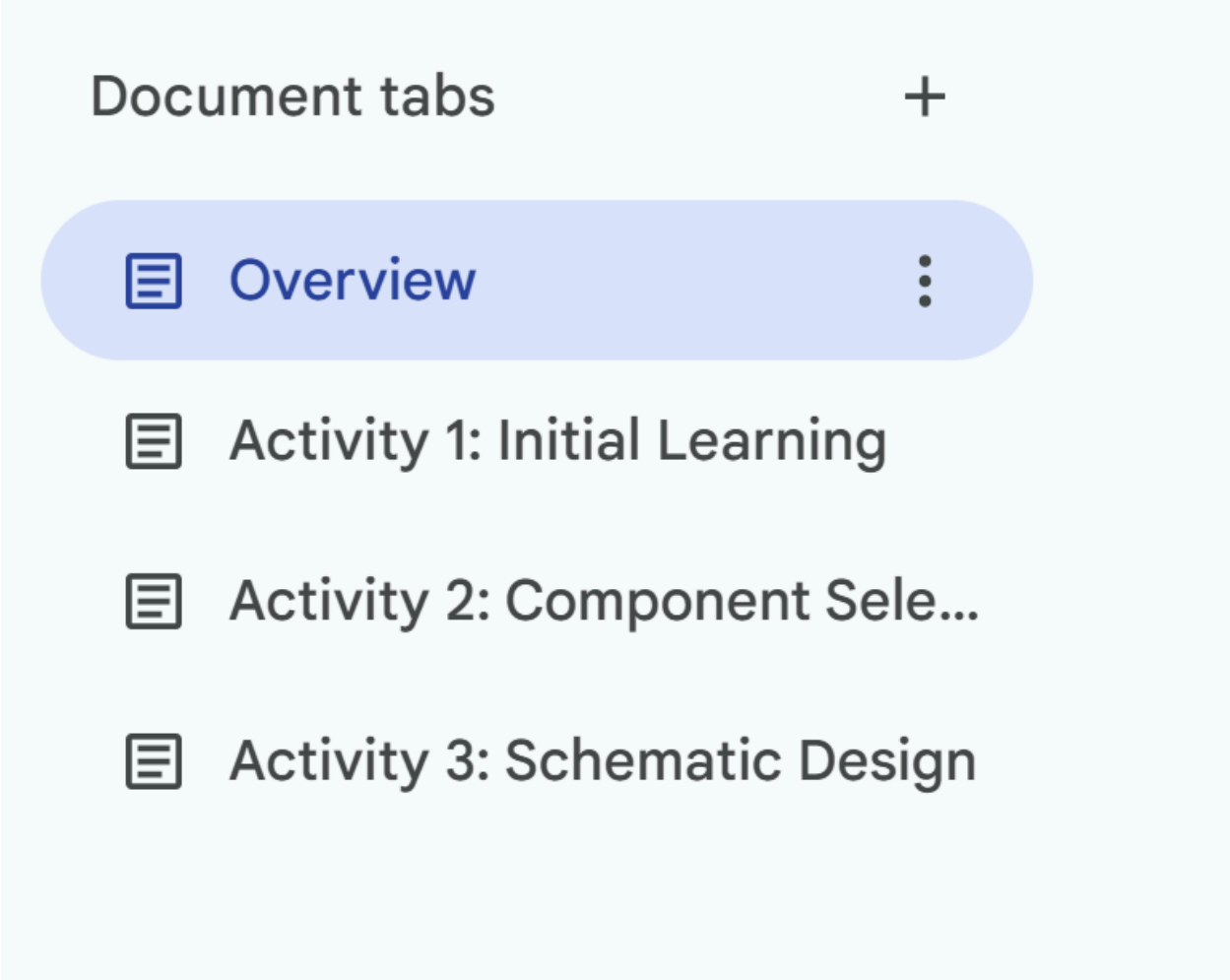
So what can we do to fix this?

A buck converter is used to transform a higher voltage to a lower voltage. We can use this to

power all of these components that require a much smaller power input, allowing it to properly function.

Navigating Project & Submission

There are 3 parts to this project: Initial Learning, Component Selection, and Schematic Layout. On the left side of the Google Doc, you will find the different tabs leading you to each activity.



Make a copy of this project, renaming it to "*Full Name* - Power Electronics Onboarding". Once every activity is completed, or before the due date, submit the document here. For example:

"Edmund Lai - Power Electronics Onboarding"

If there are any questions, message Edmund Lai on slack. There will be updates to this document if needed, check below for update logs.

Revision	Date	Updated Content
1.0	9/14/2026	Initial draft

Activity #1: Initial Learning

Listed below is a variety of initial resources that you can use if you are not familiar with electrical engineering concepts. I would suggest getting to know these well before you attempt to work on the buck converter. Any resource that talks about the same thing is good, as long as you understand the basics.

Analog Devices and Circuits:

1. Organic Chemistry Tutor: Ohm's Law
   https://www.youtube.com/watch?v=_rSHqvjDksg
2. Khan Academy, Current, Resistors, Ohm's Law:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-law/v/circuits-part-1
3. Khan Academy, Compound Circuits:
   https://www.khanacademy.org/science/ap-physics-2/x0e2f5a2c:electric-circuits/x0e2f5a2c:current-resistivity-ohms-la_w/v/electric-power
4. EEVBlog, Voltage Dividers:
   https://www.youtube.com/watch?v=xSRe_4TQbuo&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=2
5. EEVBlog, Kirchhoff's Laws:
   https://www.youtube.com/watch?v=WBfAEeEzDlg&list=PLvOlSehNtuHtWlH0UOZNtOn-FlFCn1GYw&index=13

Listed below are more power related resources. This you will need to use to understand the theory behind how switch mode power supplies work.

1. MIT OpenCourseWave Power Electronics Course:
   a. Lecture 5: Intro to DC/DC, Part 1
2. Fundamentals of Power Electronics, by Robert W. Erickson and Dragan Maksimovic
3. Power Electronics by Daniel W. Hart

Please take good notes! Take notes in your notebook, here, whatever you would want to do. I would just like to see what you guys see and how you understand it!

Current is like water but its not water
Like water if there's an easier path it goes easier path

ELECTRICITY AND MAGNETISM

$$\left|\vec{F}_E\right| = \frac{1}{4\pi\varepsilon_0}\frac{|q_1 q_2|}{r^2} = k\frac{|q_1 q_2|}{r^2}$$

$$\vec{E} = \frac{\vec{F}_E}{q}$$

$$\vec{E} = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r^2}\hat{r}$$

$$\Phi_E = \int \vec{E}\cdot d\vec{A}$$

$$\oint \vec{E}\cdot d\vec{A} = \frac{q_{\text{enc}}}{\varepsilon_0}$$

$$Q_{\text{total}} = \int \rho(r)\,dV$$

$$U_E = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r}$$

$$V = \frac{1}{4\pi\varepsilon_0}\int \frac{dq}{r}$$

$$\Delta V = -\int_a^b \vec{E}\cdot d\vec{r}$$

$$E_x = -\frac{dV}{dx}$$

$$\Delta U_E = q\Delta V$$

$$C = \frac{Q}{\Delta V}$$

$$C = \frac{\kappa\varepsilon_0 A}{d}$$

$$U_C = \frac{1}{2}Q\Delta V$$

$$\kappa = \frac{\varepsilon}{\varepsilon_0}$$

$$I = \frac{dq}{dt}$$

$$I = \int \vec{J}\cdot d\vec{A}$$

$$\vec{E} = \rho\vec{J}$$

$$R = \frac{\rho\ell}{A}$$

$$I = \frac{\Delta V}{R}$$

$$P = I\Delta V$$

A = area
C = capacitance
d = distance
E = electric field
F = force
I = current
J = current density
ℓ = length
P = power
q = charge
Q = charge
r = radius, distance, or position
R = resistance
t = time
U = potential energy
V = electric potential or volume
ε = electric permittivity
ρ = resistivity or charge density
κ = dielectric constant
Φ = flux

$$R_{\text{eq},s} = \sum_i R_i$$

$$\frac{1}{R_{\text{eq},p}} = \sum_i \frac{1}{R_i}$$

$$\frac{1}{C_{\text{eq},s}} = \sum_i \frac{1}{C_i}$$

$$C_{\text{eq},p} = \sum_i C_i$$

$$\tau = R_{\text{eq}}C_{\text{eq}}$$

$$\oint \vec{B}\cdot d\vec{A} = 0$$

$$\vec{F}_B = q\left(\vec{v}\times\vec{B}\right)$$

$$d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\left(d\vec{\ell}\times\hat{r}\right)}{r^2}$$

$$\vec{F}_B = \int I\left(d\vec{\ell}\times\vec{B}\right)$$

$$\oint \vec{B}\cdot d\vec{\ell} = \mu_0 I_{\text{enc}}$$

$$B_{\text{sol}} = \mu_0 nI$$

$$\Phi_B = \int \vec{B}\cdot d\vec{A}$$

$$\varepsilon = \oint \vec{E}\cdot d\vec{\ell} = -\frac{d\Phi_B}{dt}$$

$$\left|\varepsilon_{\text{sol}}\right| = N\left|\frac{d\Phi_B}{dt}\right|$$

$$L_{\text{sol}} = \frac{\mu_{\text{core}}N^2 A}{\ell}$$

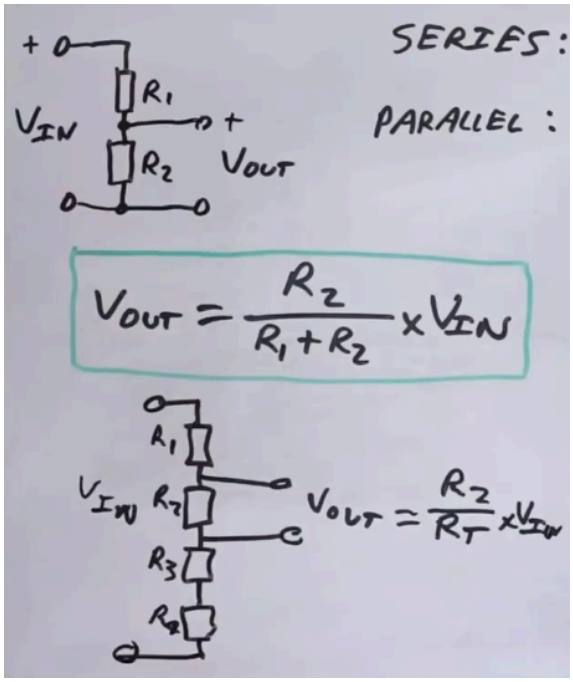
$$U_L = \frac{1}{2}LI^2$$

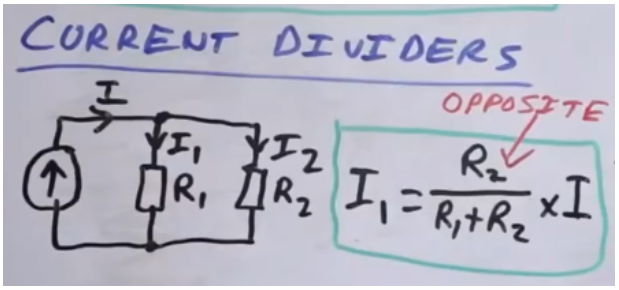
$$\varepsilon = -L\frac{dI}{dt}$$

$$\tau = \frac{L}{R_{\text{eq}}}$$

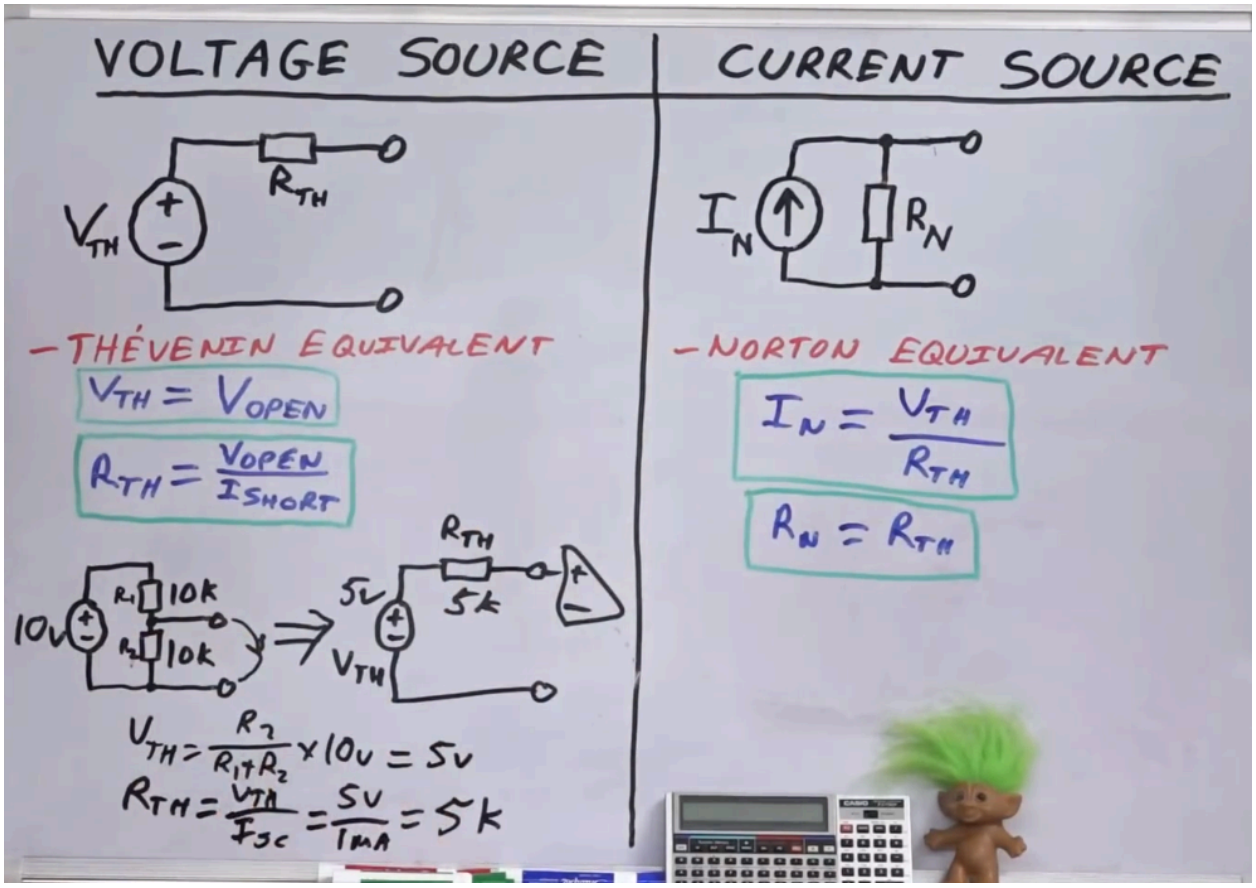
$$\omega_{LC} = \frac{1}{\sqrt{LC}}$$

A = area
B = magnetic field
C = capacitance
E = electric field
F = force
I = current
ℓ = length
L = inductance
n = number of loops per unit length
N = number of loops
q = charge
r = radius, distance, or position
R = resistance
t = time
U = potential energy
v = velocity or speed
ε = emf
μ = magnetic permeability
τ = time constant
Φ = flux
ω = angular frequency

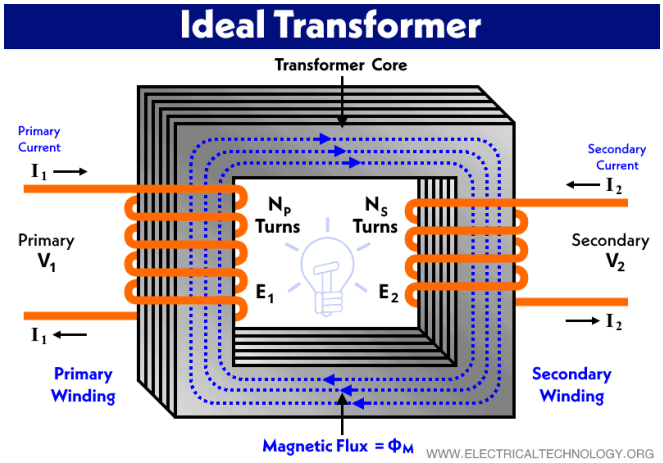


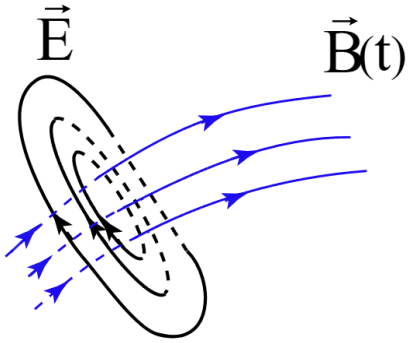


Current divider equation only applies to 2 parallel resistors



Transformer - iron core donut shape. Bars on both sides of iron cores have coils on it





- Utilizes electromagnetic induction - current produced by magnetic flux to oppose it to make magnetic flux interfering with the area enclosed by the coil
    - Current produced by induction because electric field formed around the magnetic flux
    - P = I^2 R = VI
    - Power in - heat = Power out
    - More coils have more voltage by em induction, & less current since the current's magnetic fields all stack together to oppose the same magnetic flux.
        - the more the coils the longer the path and the higher the electric potential from tip to tip? -> thats why more voltage
            - Edmund says "[iirc] voltage steps up and down due to your turns ratio because your voltage is related to the flux going through each winding and the number of windings there are "
    - for the transformer why does less coils affected by em induction mean more current? Is there more meaning to just satisfying P = IV through power transmission throughout the transformer? Less coils mean less magnetic flux over the stacked coils so im also a bit confused to whether current increases just to oppose the changing magnetic flux by em induction and if not what other things are accounted for why
- The iron core has high magnetic permeability
    - Magnetic permeability is how easy magnetic fields can pass through a material, and also how easy internal fields can be created under the effect of an external field (which explains why magnets attract to magnetic objects like iron)

PSS - periodic steady state

- Voltages & currents along a complete period are 0 because of a usage of an alternating current (AC)
    - We use AC over DC in power transmission because of how easy we can change voltage using transformers

- We up the voltage in transformers to reduce loss over power transmission by heat by current flowing in wire (P = IV)

Buh converter



Need to find peak current in the buck converter to choose an inductor

[Asynchronous Output Convertor Inductance]

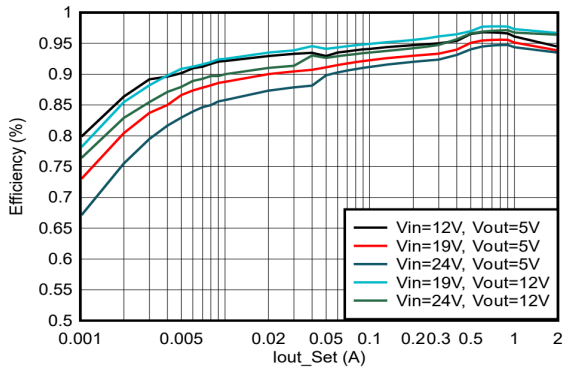
[Buck Convertor stuff]

Wow the buck ic datasheet gives the formula how generous

Use the following equation to calculate the peak inductor current ($I_{L(PK)}$).

$$I_{L(PK)} = I_{OUT(MAX)} + \frac{V_{OUT} \times \left[V_{IN(MAX)} - V_{OUT}\right]}{V_{IN(MAX)} \times L_O \times f_{SW} \times 1.6}$$

What in the tarnations is L_O?



Efficiency vs Output Current

Maximum Duty Cycle: $D = \frac{V_{OUT}}{V_{IN(max)} \times \eta}$

$V_{IN(max)}$ = maximum input voltage
V_{OUT} = output voltage
η = efficiency of the converter, e.g., estimated 90%

(buck IC efficiency)

Eyeball efficiency for Vin 12V to be ~%0.916. I'll use Vin(max) = 15V

$$\frac{5}{15(0.916666667)}$$

$$= 0.363636363504$$

Max Duty Cycle: D = 0.363636363504

$$\text{Inductor Ripple Current: } \Delta I_L = \frac{(V_{IN(max)} - V_{OUT}) \times D}{f_S \times L}$$

$V_{IN(max)}$ = maximum input voltage
V_{OUT} = desired output voltage

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D = duty cycle calculated in Equation 1
f_S = minimum switching frequency of the converter
L = selected inductor value

Nvm

If the calculated value is above the maximum output current of the application, the maximum switch current in the system is calculated:

$$\text{Application specific maximum switch current: } I_{SW(max)} = \frac{\Delta I_L}{2} + I_{OUT(max)} \quad (4)$$

ΔI_L = inductor ripple current calculated in Equation 2
$I_{OUT(max)}$ = maximum output current necessary in the application

This is the peak current, the inductor, the integrated switch(es), and the external diode have to withstand.

$$L = \frac{V_{OUT} \times (V_{IN} - V_{OUT})}{\Delta I_L \times f_S \times V_{IN}} \quad (5)$$

V_{IN} = typical input voltage
V_{OUT} = desired output voltage
f_S = minimum switching frequency of the converter
ΔI_L = estimated inductor ripple current, see the following:

The inductor ripple current cannot be calculated with Equation 1 because the inductor is not known. A good estimation for the inductor ripple current is 20% to 40% of the output current.

$$\Delta I_L = (0.2 \text{ to } 0.4) \times I_{OUT(max)} \quad (6)$$

ΔI_L = estimated inductor ripple current
$I_{OUT(max)}$ = maximum output current necessary in the application

$I_{SW(max)}=(0.3\pm 0.1)(2)/2 + 2 = 2.3\pm 0.1$ A

2

$$\frac{(5)(13.5-5)}{(0.2)(2)(500000)(13.5)} = 0.0000157407407407$$

3

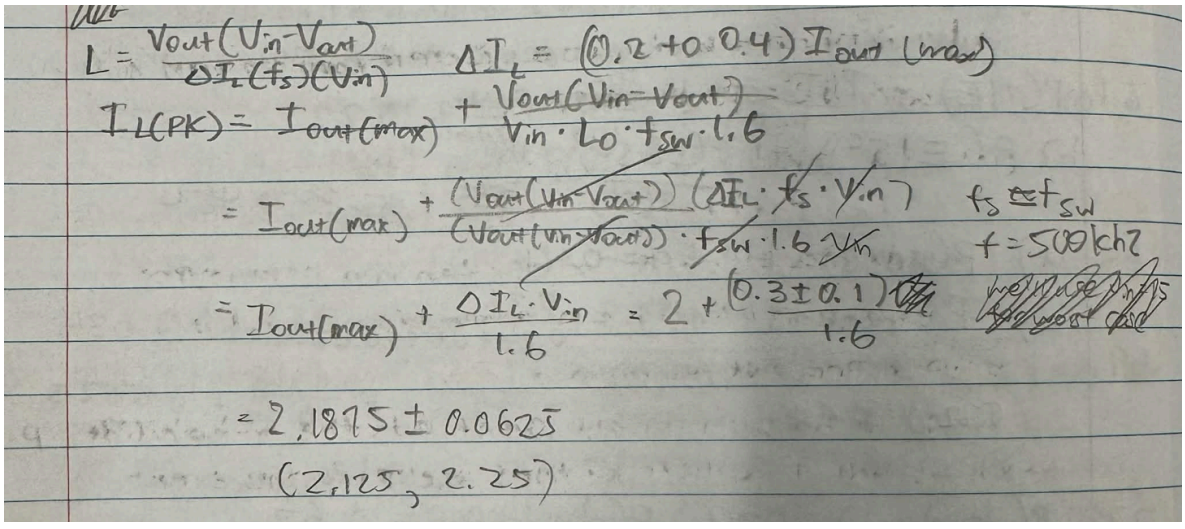
$$\frac{(5)(13.5-5)}{(0.4)(2)(500000)(13.5)} = 0.00000787037037037$$

$L_{max}=0.0000157407407407$ H
$L_{min}=0.00000787037037037$ H

Turns out L_O is indeed the inductance of the inductor
So we back to this eq

Use the following equation to calculate the peak inductor current ($I_{L(PK)}$).

$$I_{L(PK)} = I_{OUT(MAX)} + \frac{V_{OUT} \times [V_{IN(MAX)} - V_{OUT}]}{V_{IN(MAX)} \times L_O \times f_{SW} \times 1.6}$$



Let's use $I_{L(PK)} = 2.25$ A

For the output filter inductor, the RMS current and saturation current ratings must not be exceeded. Use the following equation to calculate the RMS inductor current ($I_{L(RMS)}$).

$$I_{L(MAX)} = \sqrt{I_{OUT(MAX)}^2 + \frac{1}{12} \times \left[\frac{V_{OUT} \times (V_{IN(MAX)} - V_{OUT})}{V_{IN(MAX)} \times L_O \times f_{SW} \times 0.8}\right]^2} \quad (9)$$

=sqrt($I_{out(max)}^2$+ (1/12)[(delta current subscript L)/0.8]^2)
=(2.00083315979,2.00333056018)
I'll just use $I_{L(max)}$ 2.0008A
Will look for current rating of minimum > 2.0008A
Calculate min inductance with their provided formula

Use the following equation to calculate the minimum value of the output inductor (L_{MIN}).

$$L_{MIN} = \frac{V_{OUT} \times [V_{IN(MAX)} - V_{OUT}]}{V_{IN(MAX)} \times K_{IND} \times I_{OUT} \times f_{SW}} \quad (8)$$

Where:

K_{IND} is a coefficient that represents the amount of inductor ripple current relative to the maximum output current.

In general, the value of K_{IND} is at the discretion of the designer; however, the following guidelines can be used. For designs using low-ESR output capacitors, such as ceramics, a value as high as K_{IND} = 0.3 can be used.

For this design example, use K_{IND} = 0.3. The minimum inductor value is calculated as 13.7µH. For this design, a close standard value of 15µH is selected for L_{MIN}.

L_{min} = 0.0000111111111111 = 11.1µH
There's only 10 & 12µH close value inductances to L_{min} calculated
This resource says:

You should select inductors to manage output capacitor size, load transients and output ripple current. There are benefits of both low and high inductance values.

The benefits of low inductance include:

- Lower DC resistance (DCR), which is inherent in the inductor wire, and which affects ripple and power loss.
- Higher saturation current, for higher output-current capability.
- Higher slew rate (di/dt), which improves load transient response and reduces output capacitance for a given load transient.

The benefits of high inductance include:

- Lower ripple current, which in turn reduces:
  - AC losses (inductor skin effect).
  - MOSFET root-mean-square (RMS) current.
  - Output capacitor RMS current.
  - Output capacitance for an equivalent output ripple.
  - Continuous inductor current over a wider load range.

Will just use 12µH

Wow according to source class 1 capacitors aren't affected by dc voltage bias!
Common class of class 1 capacitors is C0G

Activity #2: Component Selection

After finishing Activity #1, you are ready to start selecting different components! When selecting components, please follow the LV Standards guideline to picking different form factors. Typically, our resistors and capacitors are both 0603, but this is just a general guideline. Typically do not go under 0603, but if we need a larger voltage rating, going up is completely fine. We typically only source from Digikey as well (Mouser if it is the only place available). We also do not want to use marketplace products. These are all filters that you can apply on Digikey, and more information is available in the LV standards document.



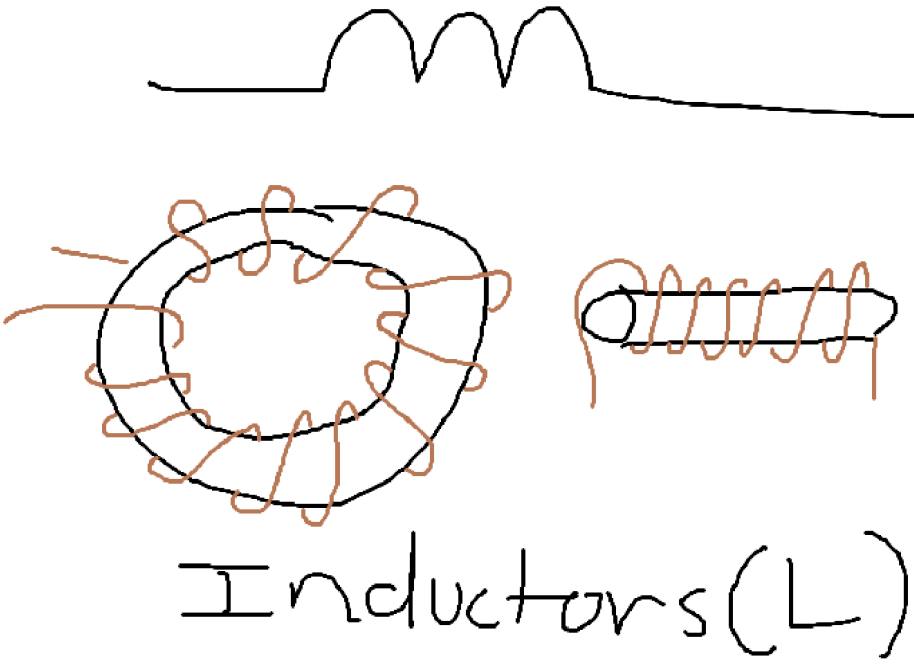
The first big thing to source is your Buck IC! This chip is used to adjust the duty cycle of the switches, and work with the feedback loop. For reference, here is the one we used last year but is now out of stock. The big thing is making sure it can output enough current to supply our boards! For FS5, we defaulted to a 2A buck, which worked perfectly fine (to my knowledge), so anything 2A and above is good! However, be sure to properly select this, because there are some tradeoffs that you may have to make if you oversize the buck converter too much. Feel free to ask me if you have any questions about that!

There are a couple I have in mind, but I would like you guys to do your own research and play with Digikey to be more familiar with the website! This is where we order almost all our parts, excluding some from Mouser. You will need to get familiar with it. Here are some initial parameters to get you all started on looking.

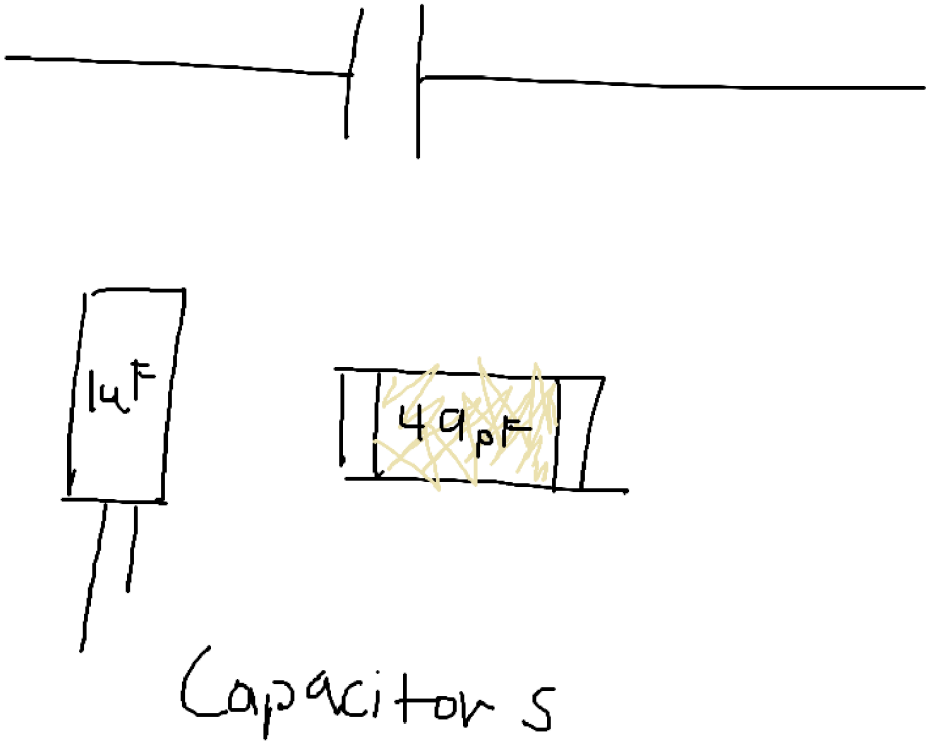
Input Voltage	~10-15V *
Output Voltage	5V
Output Current	Iout ≥ 2A
Switching Frequency	Fsw ≥ 500kHz**
Feedback Control	😳
Ripple Voltage	1-5%

*Though our LV bus is 12V nominally, it is actually 13.5V!
**with too low of a switching frequency, you upsize your inductor and output capacitor value, which could be an issue space wise. This is an arbitrary value I have set for the case of the onboarding project.

Whatever chip you decide to pick, be sure to properly read the entire thing, understanding what each pin does, what kind of internal structure it has, and the reasoning behind all of it.



One big thing you should check in an inductor is its saturation current. Make sure that your inductor does not oversaturate at the output current you are expecting, or else your inductor's inductance will drop so low that your output ripple will be extremely large.

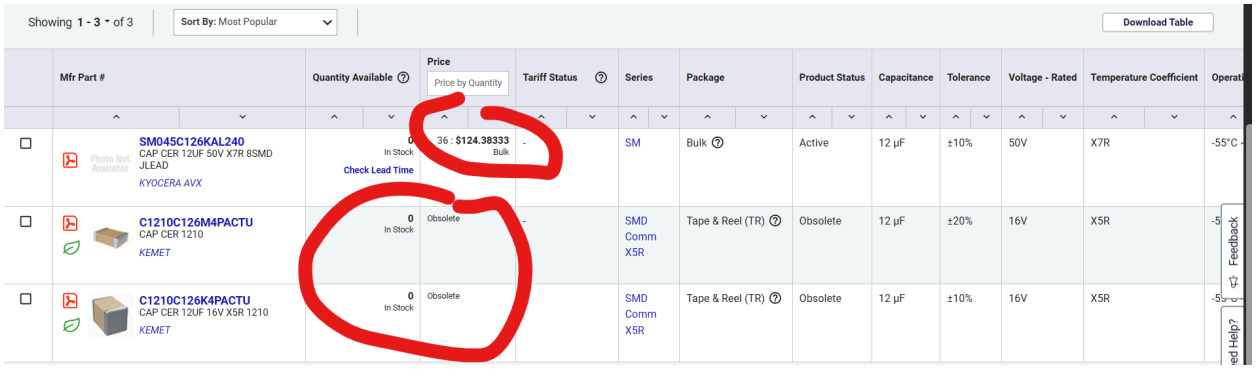


Capacitors have a funny little thing called DC Biasing. Capacitors do not like staying at the specific capacitance they were designed to be at. Instead, whenever you apply any kind of voltage across one, their capacitance will drop by some percentage, typically shown in your datasheet. Make sure that your capacitances will be what you are expecting at your expected output voltage! A little hint: The output capacitors are there to help filter noise! There is something you have to do when picking your output capacitors to help reduce noise on the output end.



Make sure to try to minimize costs as much as possible! This is more of a good engineering practice to not just go with the best thing if it costs an unreasonable amount of money.

At the end of this activity, fill out this chart with the things that you have selected for ONE buck converter, with the costs per unit and size as well. If you need to add more rows, feel free to.





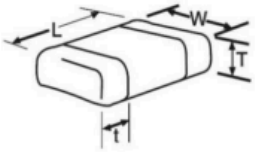
22201C106MAT2A Datasheet

(2220 100V X7R 10uF ±20%)

To download data and simulation models visit: SpiCAT ONLINE TOOL

KYOCERA AVX

Dimensions

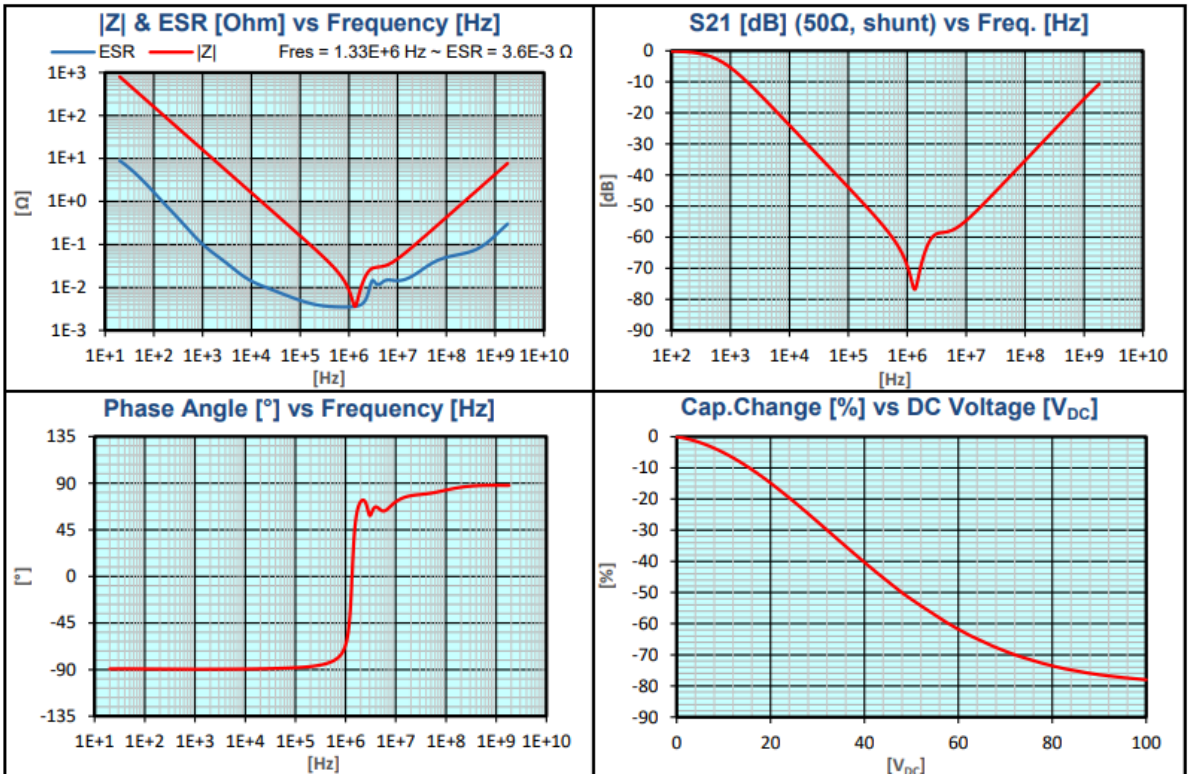


	millimetres	inches
L	5.7 ± 0.5	0.224 ± 0.02
W	5 ± 0.4	0.197 ± 0.016
T max.	2.79	0.11
t	0.64 ± 0.39	0.025 ± 0.015

Basic Specifications

Item	Unit	Spec.	Conditions
Capacitance	uF	8 to 12	@ 1 kHz, 1 Vrms
DF	%	10 max.	@ 1 kHz, 1 Vrms
IR	MΩ	100 min.	@ 100 Vdc, t = 120 s
DWV	Vdc	250	@ I ≤ 50mA, t ≤ 5 s
Operating Temperature		-55°C to +125°C	
Dielectric		X7R	
AEC-Q200		Not qualified	
RoHS Compliant		Yes	
Termination		Sn	

Electrical Characteristics



Component	Cost	Link	Units	Size
Buck IC	$1.22 (Digi- Reel order) (no tariff,	https://www.digikey.com/en/products/detail/texas-instruments/TPS	3507 in stock Buy 1	2.9mm × 2.8mm

Device Information

PART NUMBER	PACKAGE[1]	PACKAGE SIZE[2]
TPS542021	DDC (SOT-23-THN, 6)	2.9mm × 2.8mm
	DRL (SOT-563, 6)	1.60mm × 1.60mm
TPS542025	DDC (SOT-23-THN, 6)	2.9mm × 2.8mm
	DRL (SOT-563, 6)	1.60mm × 1.60mm

	manufactured by Texas Instruments)	542025DDCR/28738814		
Inductor	$1.95 (Cut Tape (CT) & Digi-Reel) (tariff applies, but in stock)	https://www.digikey.com/en/products/detail/w%C3%BCrth-elektronik/744787120/4155880	997 in stock Buy 1	7.8mm x 7mm x 5mm(height)
Capacitor 10uF	$9.88	https://www.digikey.com/en/products/detail/kyocera-avx/22201C106MAT2A/5431507	4	0.224" L x 0.197" W (5.70mm x 5.00mm)
Capacitor 100nF	$16.89	https://www.digikey.com/en/products/detail/kemet/C1210C104K5GACTU/12700257	3	0.126" L x 0.098" W (3.20mm x 2.50mm)
~~Capacitor 10nF~~	~~$0.61~~	~~https://www.digikey.com/en/products/detail/kemet/C1206C103J5GEC7210/8646242~~	~~1~~	~~0.126" L x 0.063" W (3.20mm x 1.60mm)~~
~~Capacitor 75pF~~	~~$0.42~~	~~https://www.digikey.com/en/products/detail/kyocera-avx/06031U750FAT2A/1598956~~	~~1~~	~~0.060" L x 0.030" W (1.52mm x 0.76mm)~~
Total Cost	$29.94			

https://www.digikey.com/en/products/detail/kyocera-avx/ST121C126MAJ05/2270074

So we are making an asynchronous buck converter?

Activity #3: Schematic and Layout Design

Here is where you start designing the schematic! Create a branch in the onboarding repo, and use the template KiCad project to make your buck converter. Be sure to follow the LV standards guide.

Please attend one of the KiCad workshops, where the electrical leads will teach you more about the software and what we expect.

Once you feel that your schematic and or layout is done, make a commit on the fs-electrical-onboarding repository, and push it onto github. You should make a commit no matter what, even if you did not finish completely. If you do not, you will not be considered for Power Electronics.

https://www.youtube.com/watch?v=d9_-lQq8ShE

4 Pin Configuration and Functions

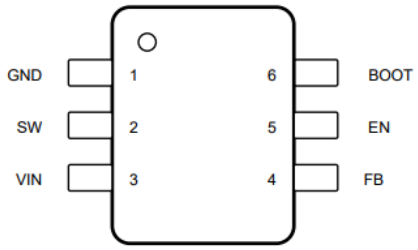


Figure 4-1. 6-Pin SOT-23 DDC Package (Top View)

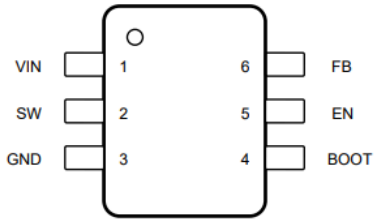


Figure 4-2. 6-Pin SOT-563 DRL Package (Top View)

Table 4-1. Pin Functions

PIN			TYPE[1]	DESCRIPTION
NAME	NO.			
	SOT-23	SOT-563		
BOOT	6	4	O	Supply input for the high-side NFET gate drive circuit. Connect a 0.1µF capacitor between BOOT and SW pins.
EN	5	5	I	Enable control for device output. Connect to 5V or leave this pin floating to enable device output. Connect to GND to disable the device output.
FB	4	6	I	Converter feedback input. Connect to output voltage with feedback resistor divider for TPS542021, connect to output capacitors to get fixed 5V output for TPS542025.
GND	1	3	—	Ground pin.
SW	2	2	O	Switch node connection between high-side NFET and low-side NFET.
VIN	3	1	—	Input voltage supply pin.

(1) O = Output; I = Input

7.2 Typical Application

7.2.1 TPS542021 5V to 30V Input, 5V Output Converter

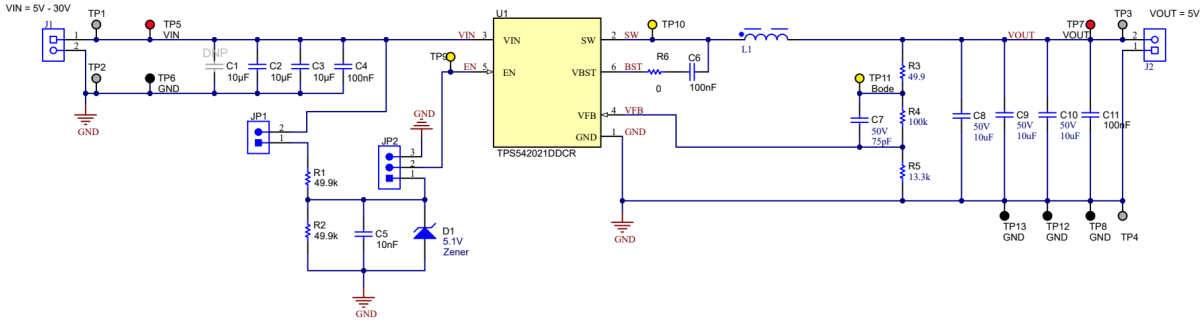
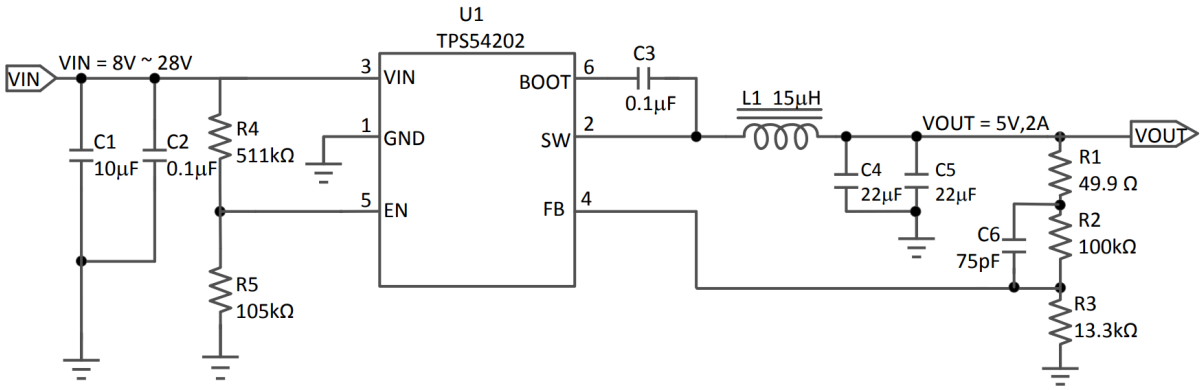


Figure 7-1. 5V, 2A Reference Design

7.2 Typical Application

7.2.1 TPS54202 8V to 28V Input, 5V Output Converter



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Extra Activity: Tractive System Shunt Current and/or Voltage Sense

If you happen to finish the buck converter, or you feel that the buck converter doesn't speak to you, you can also work on this project as well. This one will be in the form of a presentation / research project.

This project is based around adding to our current sensors within our Tractive Battery's container. Currently, we use a hall effect current sensor, which kind of looks like this:



For accuracy and data acquisition reasons, it would be nice to be able to add to this hall effect sensor with a shunt one, where you measure the voltage drop across a specially manufactured piece of metal with a specific resistance and voltage drop across, where you can then determine what current is going through it based on it. However, there are some constraints to this system, ie: isolation, heat dissipation, etc

Our current tractive battery current can go up to ~600A peak. We want to be able to measure up to this and down to the negatives as well in order to track regen current. Required by FSAE 26-27 Rules (EV.8.7.4), we need to be able to *prove* to judges that the Brake System Plausibility Device works through simulating tractive system current going through the current sensor. Find a way to be able to simulate this using a shunt if possible. At the moment, we have wire wrapped around a hall effect sensor multiple times, simulating higher currents with a smaller test current.

We would also like to be able to properly measure the voltage in our tractive system while we drive. We should find a way to take that voltage measurement, isolate it and pass it through to a

microcontroller so that we can have this data for the future.

When completing this research project, continue to write good documentation and make sure to optimize for costs when possible.

Have fun!